Binary Neutron Stars

Day 2: Dense Matter Equation of State, Anisotropic Pressure, and Superfluidity

Comprehensive Lecture Notes with Detailed Discussion

Three major questions today:

I. What determines the neutron star EOS at supranuclear density?

II. How does relaxing isotropy modify stellar structure?

III. How does superfluidity modify internal dynamics?

These notes expand on the Day 2 slides. Boxed material highlights key results, physical interpretation, subtle points, common misconceptions, and advanced topics. References given at the end.

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1 Overview and Conceptual Roadmap

In Day 1 we derived the Tolman–Oppenheimer–Volkoff (TOV) equation, which determines neutron star structure given the equation of state:

$$\frac{dp}{dr} = -\frac{(\epsilon + p)(m + 4\pi r^3 p)}{r(r - 2m)}, \qquad \frac{dm}{dr} = 4\pi r^2 \epsilon. \tag{1}$$

Today we open the black box on the right-hand side: the EOS $p = p(\epsilon)$. We also extend the TOV framework in two directions — allowing **anisotropic pressure** and introducing **superfluidity**.

Every topic we cover today feeds into something observable:

- **EOS stiffness and phase transitions** → mass–radius relation, maximum mass, tidal deformability Λ.
- **Symmetry energy slope** L → neutron star radius (the most sensitive single-parameter correlation).
- **Anisotropic pressure** → modified TOV equilibrium, altered Love numbers.
- **Superfluidity** → oscillation mode spectrum, pulsar glitches, dynamical tidal response.

Understanding the microphysics is not academic — each piece has a direct signature in gravitational-wave observations of binary mergers.

2 Cold Catalyzed Matter

2.1 Why "Cold"?

Neutron stars are extraordinarily cold *compared to the relevant energy scales.* The Fermi energy of the baryons inside a neutron star is

$$E_F \sim 10\text{–}100 \text{ MeV}. \tag{2}$$

A few minutes after formation, the interior temperature has already dropped to

$$T \lesssim 10^{10} \text{ K} \sim 1 \text{ MeV}, \tag{3}$$

and within $\sim 10^4$ yr it cools further to

$$T \sim 10^8 \text{ K} \sim 10^{-2} \text{ MeV}. \tag{4}$$

Thus during the inspiral phase of a binary merger (which is what gravitational-wave detectors observe), the neutron stars satisfy

$$T \ll E_F. \tag{5}$$

Thermal corrections to the EOS are of order $(T/E_F)^2 \ll 1$ and are safely neglected.

The cold EOS is excellent for the inspiral and for isolated neutron stars. It breaks down for:

- The **merger remnant**: the post-merger disk and hypermassive neutron star

can reach $T \sim 50$–100 MeV. Thermal pressure modifies both the EOS and the stability of the remnant. The post-merger gravitational-wave signal (at frequencies ~ 2–4 kHz) requires a finite-temperature EOS.

- **Newly born proto-neutron stars** (first ~ 60 s post-bounce): trapped neutrinos and high T both need to be included.

2.2 What Does "Catalyzed" Mean?

"Cold catalyzed matter" means matter in its absolute ground state at each density, with all weak-interaction processes having reached equilibrium. Concretely this means:

1. **Beta equilibrium**: weak interactions have equilibrated, so the composition (proton fraction, lepton fractions) is determined by the minimum-energy state at each density.
2. **Charge neutrality**: no net charge is carried by the bulk matter.
3. **Baryon number conservation**: total baryon number is fixed.

Under these conditions, the entire thermodynamic state of the matter at $T = 0$ is determined by a single quantity: the **baryon number density** n. Everything else — pressure, energy density, composition — follows from the EOS.

In the bulk of the neutron star core, yes: the weak interaction timescale is much shorter than any relevant dynamical timescale (except during the very fastest oscillations or GW-driven inspirals). However, in the **crust**, matter is not in full equilibrium: the lattice of neutron-rich nuclei is in a metastable state locked in by the Coulomb energy. True catalysis would require the nuclei to dissolve into a uniform liquid, which is kinetically suppressed at low temperature. This distinction matters for calculating crust properties (shear modulus, breaking strain) relevant to pulsar glitches.

3 Thermodynamics at Zero Temperature

3.1 First Law at $T = 0$

The thermodynamic identity for matter at zero temperature reduces to:

$$d\epsilon = \mu\, dn, \tag{6}$$

where μ is the **baryon chemical potential** (energy per baryon, including rest mass). This is the $T = 0$ limit of the full first law $d\epsilon = T\, ds + \mu\, dn$ with $ds = 0$.

From the Euler relation for a homogeneous system:

$$\epsilon + p = \mu n, \tag{7}$$

which gives the **pressure** as:

$$p = n\mu - \epsilon. \tag{8}$$

Deriving $p = n\mu - \epsilon$ from thermodynamics

Consider a volume V containing N baryons. The total energy is $E = \epsilon V$ and baryon number is $N = nV$. At constant N and $T = 0$, the pressure is:

$$p = -\left.\frac{\partial E}{\partial V}\right|_{N,S} = -\left.\frac{\partial(\epsilon V)}{\partial V}\right|_{N} = -\epsilon - V\frac{d\epsilon}{dn}\left.\frac{\partial n}{\partial V}\right|_{N}.$$

At fixed N, changing V changes $n = N/V$:

$$\left.\frac{\partial n}{\partial V}\right|_N = -\frac{n}{V}, \qquad \frac{d\epsilon}{dn} = \mu.$$

Substituting:

$$p = -\epsilon + n\mu = n\mu - \epsilon.$$

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This can also be written as the Gibbs–Duhem relation at $T = 0$: $dp = n\,d\mu$.

3.2 Sound Speed

Taking the derivative of Eq. (8) with respect to n:

$$\frac{dp}{dn} = \mu + n\frac{d\mu}{dn} - \frac{d\epsilon}{dn} = \mu + n\frac{d\mu}{dn} - \mu = n\frac{d\mu}{dn}. \tag{9}$$

The squared speed of sound is therefore:

$$c_s^2 = \frac{dp}{d\epsilon} = \frac{dp/dn}{d\epsilon/dn} = \frac{n\,d\mu/dn}{\mu} = \frac{n}{\mu}\frac{d\mu}{dn}. \tag{10}$$

The speed of sound c_s governs how quickly pressure perturbations propagate through the neutron star interior. It determines:

- The **stiffness** of the EOS: large c_s^2 means pressure rises steeply with density (stiff).
- The **stability** of oscillation modes: imaginary c_s signals a thermodynamic instability (phase transition).
- The **speed of the normal fluid**: in a superfluid, both first and second sound speeds are relevant.

3.3 Constraints on the Sound Speed

Two fundamental constraints bound c_s^2:

1. **Thermodynamic stability** ($dp/d\epsilon > 0$): If $c_s^2 < 0$, a density fluctuation grows exponentially — the matter is unstable and will phase-separate. This is the signature of a first-order phase transition (where the matter separates into two coexisting phases).

2. **Causality** ($c_s \leq c$, i.e. $dp/d\epsilon \leq 1$): The speed of sound cannot exceed the speed of light. In geometrised units this is $c_s^2 \leq 1$.

In conformal field theory (massless, scale-invariant matter), the speed of sound is exactly

$$c_s^2 = \frac{1}{3}. \tag{11}$$

This is the value for an ultra-relativistic gas or for perturbative QCD at asymptotically high density. However, **at intermediate densities** ($n \sim 2$–$5\,n_0$), current nuclear theory and astrophysical constraints together indicate that c_s^2 likely exceeds $1/3$ — sometimes

significantly. This **violation of the conformal limit** is expected physically: strong repulsive nuclear interactions at intermediate density stiffen the matter beyond what pure kinetic energy arguments suggest. This is sometimes called the "peak" in the sound speed profile, and its location and height encode information about possible phase transitions or quark matter onset.

4 Degenerate Fermi Gas: The Baseline

We denote by n_0 the nuclear saturation number density, i.e. the equilibrium density of infinite nuclear matter:

$$n_0 \approx 0.16 \text{ fm}^{-3}.$$

4.1 Free Relativistic Fermions

For a single species of fermion with mass m and Fermi momentum k_F:

Fermi sphere and number density

At $T = 0$, fermions occupy all momentum states up to k_F, forming a **Fermi sphere** in momentum space.
Each quantum state occupies a phase space volume:

$$(2\pi\hbar)^3.$$

Including spin degeneracy $g = 2$, the number density is:

$$n = g \int_{|\boldsymbol{k}| \le k_\mathrm{F}} \frac{d^3k}{(2\pi\hbar)^3} = \frac{2}{(2\pi\hbar)^3} \cdot \frac{4\pi}{3} k_\mathrm{F}^3 = \boxed{\frac{k_\mathrm{F}^3}{3\pi^2\hbar^3}}. \tag{12}$$

In natural units ($\hbar = 1$):

$$n = \frac{k_\mathrm{F}^3}{3\pi^2}.$$

The relation $n \propto k_\mathrm{F}^3$ is purely geometric: increasing density forces fermions into higher momentum states due to the Pauli exclusion principle.

$$\epsilon = \frac{1}{\pi^2\hbar^3} \int_0^{k_\mathrm{F}} \sqrt{k^2c^2 + m^2c^4}\, k^2 dk, \tag{13}$$

$$p = \frac{1}{3\pi^2\hbar^3} \int_0^{k_\mathrm{F}} \frac{k^4c^2}{\sqrt{k^2c^2 + m^2c^4}}\, dk. \tag{14}$$

Origin of the pressure integrand

The pressure follows from the spatial components of the energy–momentum tensor:

$$T^{ij} = \int \frac{d^3k}{(2\pi\hbar)^3} \frac{k^i k^j}{E}. \tag{15}$$

For an isotropic system:

$$T^{ij} = p\,\delta^{ij} \quad \Rightarrow \quad p = \frac{1}{3}\int \frac{d^3k}{(2\pi\hbar)^3}\frac{k^2}{E}.$$

Using $d^3k = 4\pi k^2 dk$, we obtain:

$$p \propto \int \frac{k^2}{E}\, k^2 dk = \int \frac{k^4}{E} dk.$$

This explains the structure of the integrand

$$\frac{k^4}{\sqrt{k^2c^2 + m^2c^4}}.$$

Pressure is **degeneracy pressure**: it exists even at $T = 0$ and is dominated by high-momentum states near the Fermi surface.

Always keep track of $(2\pi\hbar)^3$ in phase space. In neutron star physics, one typically sets $\hbar = c = 1$.

4.2 Free Relativistic Fermions

Before including nuclear interactions, it is useful to work out the free Fermi gas, which provides the baseline against which interactions are measured.

For a single species of fermion with mass m and Fermi momentum k_{F}:

$$n = \frac{k_{\mathrm{F}}^3}{3\pi^2}, \tag{16}$$

$$\epsilon = \frac{1}{\pi^2}\int_0^{k_{\mathrm{F}}} \sqrt{k^2 + m^2}\; k^2\, dk, \tag{17}$$

$$p = \frac{1}{3\pi^2}\int_0^{k_{\mathrm{F}}} \frac{k^4}{\sqrt{k^2+m^2}}\, dk. \tag{18}$$

Non-relativistic and ultra-relativistic limits

Non-relativistic limit ($k_{\mathrm{F}} \ll m$):
Expand $\sqrt{k^2+m^2} \approx m + k^2/(2m)$:

$$\epsilon \approx mn + \frac{3}{5}\frac{k_{\mathrm{F}}^2}{2m} n = mn\left(1 + \frac{3}{10}\frac{k_{\mathrm{F}}^2}{m^2}\right),$$

$$p \approx \frac{k_{\mathrm{F}}^2}{3m}\cdot\frac{n}{3} = \frac{k_{\mathrm{F}}^2 n}{5m} \propto n^{5/3}.$$

The pressure is purely kinetic (degeneracy pressure) and scales as $p \propto n^{5/3}$, corresponding to a polytropic index $\Gamma = 5/3$.
Ultra-relativistic limit ($k_F \gg m$):
Expand $\sqrt{k^2 + m^2} \approx k$:

$$\epsilon \approx \frac{1}{\pi^2} \int_0^{k_F} k^3 \, dk = \frac{k_F^4}{4\pi^2},$$
$$p \approx \frac{1}{3\pi^2} \int_0^{k_F} k^3 \, dk = \frac{k_F^4}{12\pi^2} = \frac{\epsilon}{3}.$$

In the ultra-relativistic limit, $p = \epsilon/3$, which is the equation of state of radiation (or massless particles). The polytropic index is $\Gamma = 4/3$.

4.3 Why the Free Gas is Insufficient

Neutron star matter is *not* a free Fermi gas. At saturation density $n_0 \approx 0.16$ fm^{-3}, the inter-particle spacing is ~ 1.5 fm, comparable to the range of the nuclear force ($\sim$ 1–2 fm). Strong interactions are therefore not a perturbation but a dominant contribution.

Quantitatively: the binding energy of symmetric nuclear matter at saturation is

$$E_{\text{bind}}/A = -16 \text{ MeV}, \tag{19}$$

meaning nucleons are *16 MeV lower in energy* than free nucleons. This is a 1.7% correction to the rest mass energy ($m_n c^2 \approx 939$ MeV) but a $\sim 100\%$ correction to the *kinetic* energy at saturation. Nuclear interactions completely reshape the EOS.

5 Beta Equilibrium and Charge Neutrality

5.1 The Composition of Neutron Star Matter

At densities above $\sim n_0$, the relevant particle species are neutrons (n), protons (p), electrons (e^-), and muons (μ^-) (with muons appearing above ~ 0.12 fm^{-3}, when $\mu_e \geq m_\mu c^2 = 105$ MeV). At higher densities, hyperons, meson condensates, or quark matter may also appear.

5.2 Beta Equilibrium

The relevant weak processes are:

$$n \to p + e^- + \bar{\nu}_e, \tag{20}$$
$$p + e^- \to n + \nu_e. \tag{21}$$

In bulk stellar matter, neutrinos escape freely (for cold, mature NS), so the equilibrium condition is obtained by setting $\mu_{\nu_e} = 0$:

$$\mu_n = \mu_p + \mu_e. \tag{22}$$

For muons, the analogous condition gives $\mu_\mu = \mu_e$.

Equation (22) is the condition that converting a neutron to a proton plus electron (or vice versa) costs zero free energy. It constrains the proton fraction $x_p = n_p/n$: at low density, $x_p \approx 0.04$–0.05 (very neutron-rich); at high density, x_p increases toward ~ 0.10–0.20 as the symmetry energy grows.

5.3 Charge Neutrality

In the ground state, bulk matter must be electrically neutral. For a $npe\mu$ system:

$$n_p = n_e + n_\mu. \tag{23}$$

Without this condition, the Coulomb energy density would diverge as $\sim n_p^2 R^2$ for a star of radius R — enormously large. Charge neutrality is enforced by the long-range Coulomb interaction on scales larger than the Debye screening length ($\sim$ fm in dense matter).

5.4 Consequences for Cooling: The Direct URCA Process

The proton fraction determines whether the rapid **direct URCA** cooling process operates:

$$n \to p + e^- + \bar{\nu}_e, \quad p + e^- \to n + \nu_e. \tag{24}$$

Direct URCA is only kinematically allowed if momentum conservation can be satisfied, which requires:

$$k_{\mathrm{F}}^{(p)} + k_{\mathrm{F}}^{(e)} \geq k_{\mathrm{F}}^{(n)}, \tag{25}$$

equivalent to a threshold proton fraction. For a pure npe system:

$$x_p^{\mathrm{URCA}} \approx \frac{1}{9} \approx 11\%.$$

In realistic neutron star matter including muons ($npe\mu$), the threshold is higher,

$$x_p^{\mathrm{URCA}} \approx 0.13\text{–}0.15,$$

since electrons carry only part of the charge density.

Beta equilibrium does *not* mean that weak interactions stop. Rather, forward and reverse reactions occur at equal rates:

$$n \leftrightarrow p + e^- + \bar{\nu}_e.$$

Even in equilibrium, each reaction emits a neutrino or antineutrino that escapes the star, carrying away energy. Thus:

- Net composition remains constant (equilibrium),
- Energy is continuously lost via neutrino emission (cooling).

Equilibrium refers to *chemical balance*, not to the absence of microscopic processes.

If direct URCA operates, neutron stars cool by ~ 6 orders of magnitude faster than via the modified URCA process. Rapid cooling affects the thermal emission spectrum and, more subtly, the **bulk viscosity** of the neutron star matter. Bulk viscosity damps oscillation

modes, which modifies the dynamical tidal response in the late inspiral. Whether direct URCA is active depends on the symmetry energy — so cooling observations and tidal measurements are complementary probes of the same nuclear physics.

6 Nuclear Symmetry Energy

The strong interaction is approximately invariant under exchange of neutrons and protons. This motivates the concept of **isospin**, in which protons and neutrons are treated as two states of the same particle (the nucleon).
In this language:

$$p \sim \left|+\tfrac{1}{2}\right\rangle, \qquad n \sim \left|-\tfrac{1}{2}\right\rangle.$$

Isospin symmetry is only approximate, being broken by electromagnetic interactions and the neutron–proton mass difference.

6.1 The Symmetry Energy Framework

Here A denotes the total baryon number, so that E/A is the energy per baryon.

The energy per baryon of uniform nuclear matter can be expanded around isospin-symmetric matter ($n_n = n_p$, i.e. isospin asymmetry $\delta = 0$):

$$\frac{E}{A}(n,\delta) = \frac{E}{A}(n,0) + S(n)\,\delta^2 + \mathcal{O}(\delta^4), \tag{26}$$

where

$$\delta = \frac{n_n - n_p}{n} \tag{27}$$

is the isospin asymmetry parameter. The function $S(n)$ is the **density-dependent symmetry energy**. Odd powers of δ vanish by isospin symmetry.

Pure neutron matter (neutron stars) has $\delta \approx 1$ (nearly), while symmetric nuclear matter has $\delta = 0$.

The symmetry energy $S(n)$ quantifies the energy cost of converting protons into neutrons, i.e. of making matter neutron-rich. Equivalently, it measures how much energy is required to deviate from isospin-symmetric matter ($n_n = n_p$).

6.2 Key Parameters at Saturation Density

It is conventional to expand $S(n)$ around n_0:

$$S(n) = J + L\,\frac{n-n_0}{3n_0} + \frac{K_{\rm sym}}{2}\left(\frac{n-n_0}{3n_0}\right)^2 + \dots \tag{28}$$

where:

$$J = S(n_0) \approx 30\text{–}35 \text{ MeV} \quad \text{(symmetry energy at saturation)}, \tag{29}$$

$$L = 3n_0 \left.\frac{dS}{dn}\right|_{n_0} \approx 40\text{–}80 \text{ MeV} \quad \text{(slope parameter)}. \tag{30}$$

Why does L control the pressure?

The pressure is:

$$p = n^2 \frac{d}{dn}\left(\frac{E}{A}\right). \tag{31}$$

Using $\frac{E}{A} = \frac{E}{A}(n, 0) + S(n)\delta^2$ and $\delta \approx 1$ for neutron star matter:

$$p_{\text{sym}} \sim n^2 \frac{dS}{dn}. \tag{32}$$

At n_0, $\frac{dS}{dn} = \frac{L}{3n_0}$, hence:

$$p_{\text{sym}} \sim n^2 \frac{L}{3n_0}. \tag{33}$$

Thus larger L implies larger pressure.

The slope parameter L is the single nuclear-physics quantity most strongly correlated with neutron star radius. A larger L means the symmetry energy increases more steeply with density above n_0, providing more pressure in the neutron-rich outer core. A larger L means $S(n)$ rises more rapidly with density, making neutron–proton imbalance increasingly costly and generating more pressure in the neutron-rich outer core ($n \sim n_0$–$2n_0$). Empirically:

$$R_{1.4} \approx a + b\, L^{1/4}, \tag{34}$$

where $R_{1.4}$ is the radius of a $1.4\, M_\odot$ star. The correlation is remarkably tight across diverse EOS models. Current constraints from GW170817 + NICER give $L \approx 40$–80 MeV.

Why does L control R? The outer core of a neutron star ($n \sim n_0$–$2n_0$) is where most of the volume (and hence the radius) is determined. In this region, the pressure is dominated by the neutron–proton asymmetry, i.e. the symmetry energy. A larger L means the symmetry pressure rises faster with density, "puffing out" the star to a larger radius. By contrast, the maximum mass is controlled by the EOS at much higher densities ($n \sim 3$–$7\, n_0$), where a different set of parameters dominates. This explains why M_{max} and $R_{1.4}$ are (somewhat) independent probes of nuclear physics.

6.3 Laboratory Constraints on the Symmetry Energy

The symmetry energy parameters are not just astrophysical — they are also measured in nuclear experiments:

- **Nuclear masses and binding energies**: constrain J and the bulk surface symmetry energy.
- **Neutron skin thickness of ^{208}Pb**: The PREX-II experiment measured the neutron skin thickness $\Delta r_{np} = 0.283 \pm 0.071$ fm, favouring a large $L \approx 106 \pm 37$ MeV. However, this is in mild tension with GW170817 constraints.

- **Giant and pygmy dipole resonances**: probe J and L via collective excitations.
- **Heavy-ion collisions**: probe the symmetry energy at sub- and supra-saturation densities.

The PREX-II result for L is significantly larger than most astrophysical constraints. This tension is not yet resolved. Possible explanations include: systematic errors in the PREX-II extraction, new physics in the neutron skin, or genuine inconsistencies between terrestrial and astrophysical measurements that hint at beyond-standard-model nuclear physics. The resolution will require both improved nuclear experiments and more precise GW/X-ray measurements.

7 Phase Transitions in Dense Matter

7.1 First-Order Phase Transitions

Two phases coexist at the same pressure and chemical potential, but have different energy (and number) densities. In a phase coexistence region, the pressure is constant while the energy density jumps:

$$p = p_{\text{trans}} = \text{const}, \qquad \epsilon_1 < \epsilon < \epsilon_2. \tag{35}$$

The EOS has a "horizontal segment" in the p–ϵ plane, corresponding to $c_s^2 = dp/d\epsilon = 0$.

At the phase transition, matter converts between Phase 1 (lower density) and Phase 2 (higher density) at constant pressure. In a neutron star, this produces a **mixed phase region** where both phases coexist — unless surface tension is large, in which case a sharp **phase boundary** forms instead.
The effect on the EOS is a softening: the pressure does not increase during the transition, so the star cannot support as much weight. This typically lowers the maximum mass.

7.2 Consequences: Twin Stars

One of the most dramatic consequences of a first-order phase transition is the possible appearance of **twin stars** (also called "third family" compact stars).

Consider an EOS with:

1. A hadronic phase at low density,
2. A strong first-order transition at some n_{trans},
3. A stiff high-density phase (e.g. quark matter).

The M–R diagram can develop *two separate stable branches*:

- A hadronic branch (conventional neutron stars),
- A hybrid branch (quark-core stars).

Two stars with identical masses but very different radii become possible — hence "twin stars".

If twin stars exist, a binary merger could in principle involve one star from each branch. The tidal deformabilities of the two stars would be very different, producing a distinctive asymmetry in the gravitational-wave signal. The post-merger remnant could also collapse

promptly to a black hole if the heavier component is on the hadronic branch but would survive as a neutron star if both components are on the hybrid branch. Future detectors with sensitivity to the $\sim$ 2–4 kHz post-merger band could detect these signatures.

7.3 Second-Order Phase Transitions

A second-order (continuous) phase transition has no latent heat and no density discontinuity. The EOS is continuous but its *derivatives* can be discontinuous. The sound speed $c_s^2 = dp/d\epsilon$ may show a kink.

Examples include:

- The onset of superfluidity/superconductivity (pairing transition),
- The onset of pion or kaon condensation (in some models),
- Chiral symmetry restoration.

7.4 Quark Matter

At asymptotically high density, QCD is perturbatively tractable and the ground state is a colour superconductor (quarks form Cooper pairs). At intermediate densities ($n \sim 3$–$10\, n_0$), the situation is far less clear.

At sufficiently high densities, nucleons are expected to dissolve into their constituent quarks. Instead of baryons as the relevant degrees of freedom, the system is described by deconfined quarks (typically u, d, and possibly s) forming a Fermi sea.
In this regime, the pressure arises primarily from quark degeneracy pressure and interactions governed by QCD. Compared to hadronic matter, quark matter can be either softer or stiffer depending on the strength of interactions and pairing effects.

The most studied possibility is the **colour-flavour locked (CFL) phase**: all three quark flavours (u, d, s) pair symmetrically, breaking colour and flavour symmetry down to a diagonal subgroup. The CFL phase has:

- A large gap $\Delta \sim 50$–150 MeV,
- Baryon superfluidity,
- A stiff EOS (if the gap and bag constant are right).

The QCD phase diagram in the temperature–baryon chemical potential (T–μ_B) plane has at least two well-established features: the hadronic phase at low μ_B and the quark–gluon plasma at high T. Heavy-ion colliders (RHIC, LHC) probe the high-T, low-μ_B regime. Neutron stars probe the low-T, high-μ_B regime — the opposite corner. Whether these two extreme regions of the phase diagram are connected by a single first-order transition, a crossover, or some more complex topology is one of the central unsolved problems of QCD. Gravitational-wave observations of NS mergers are the only laboratory for the low-T, high-μ_B regime, which is inaccessible to any particle accelerator.

8 Anisotropic Pressure: Theory

8.1 Motivation and Physical Sources

The TOV equation assumes **isotropic pressure**: $p_r = p_t$, where p_r is the radial pressure and p_t is the tangential pressure. This assumption can be violated in several physically motivated scenarios:

1. **Crust elasticity**: The solid neutron star crust has a non-zero shear modulus $\mu_{\text{crust}} \sim 10^{30}$–$10^{31}$ erg cm^{-3}. Elastic stresses produce $p_r \neq p_t$ in the crust. However, the crust constitutes only $\sim$ 1–2% of the star's volume and mass, so the bulk effect is small.
2. **Strong magnetic fields**: A magnetic field B introduces a pressure anisotropy $\sim B^2/(8\pi)$. For magnetars with $B \sim 10^{15}$ G, this gives $p_{\text{mag}} \sim 4 \times 10^{31}$ erg cm^{-3}, which is comparable to the matter pressure in the outer layers and breaks spherical symmetry.
3. **Superfluid vortex lattice**: The rotation of the superfluid induces a vortex lattice that contributes anisotropic stress.
4. **Anisotropic solid phases**: If dense matter forms an anisotropic crystalline structure (e.g. nuclear pasta in the inner crust, or a crystalline colour superconductor in quark matter), the stress tensor is inherently anisotropic.
5. **Phenomenological models**: Independent of microphysical motivation, one can parameterise anisotropy (following Bowers & Liang 1974) to study its structural effects.

8.2 The Anisotropic Stress-Energy Tensor

For a spherically symmetric but pressure-anisotropic fluid, the stress-energy tensor takes the form:

$$T^{\mu}_{\ \nu} = \text{diag}(-\epsilon,\ p_r,\ p_t,\ p_t), \tag{36}$$

with two independent pressure functions $p_r(r)$ and $p_t(r)$. The anisotropy parameter is:

$$\sigma(r) = p_t(r) - p_r(r). \tag{37}$$

When $\sigma = 0$ the isotropic case is recovered.

A purely radial pressure p_r and a purely tangential pressure p_t are both compatible with spherical symmetry, as long as p_r and p_t depend only on r. Spherical symmetry does not require isotropy — it only requires that the stress tensor commutes with the generators of rotations. The diagonal form $T^{\mu}_{\ \nu} = \text{diag}(-\epsilon, p_r, p_t, p_t)$ satisfies this precisely.

9 Derivation of the Anisotropic TOV Equation

9.1 Energy-Momentum Conservation

The anisotropic TOV is derived from $\nabla_\mu T^{\mu\nu} = 0$, exactly as in the isotropic case, but now with the more general stress tensor.

The r-component of $\nabla_\mu T^{\mu r} = 0$ gives:

$$\frac{dp_r}{dr} = -\frac{1}{2}(\epsilon + p_r)\,\nu' + \frac{2}{r}(p_t - p_r). \tag{38}$$

The first term is the usual isotropic TOV structure. The second term is the **anisotropy force**.

Derivation of the anisotropy term

The covariant divergence of $T^{\mu\nu}$ for the metric $ds^2 = -e^\nu dt^2 + e^\lambda dr^2 + r^2 d\Omega^2$ gives for the r-component:

$$\partial_r T^r{}_r + \Gamma^r{}_{rr} T^r{}_r - \Gamma^r{}_{tt} T^t{}_t - 2\Gamma^\theta{}_{r\theta} T^\theta{}_\theta = 0.$$

The relevant Christoffel symbols are:

$$\begin{aligned} \Gamma^r{}_{rr} &= \tfrac{1}{2}\lambda', \\ \Gamma^r{}_{tt} &= \tfrac{1}{2}\nu' e^{\nu-\lambda}, \\ \Gamma^\theta{}_{r\theta} &= \frac{1}{r}. \end{aligned}$$

Substituting $T^r{}_r = p_r$, $T^t{}_t = -\epsilon$, $T^\theta{}_\theta = p_t$:

$$p_r' + \tfrac{1}{2}\lambda' p_r + \tfrac{1}{2}\nu' e^{\nu-\lambda}(-\epsilon)e^{-\nu+\lambda} - \frac{2}{r}p_t = 0.$$

Rearranging and using $e^{-\lambda} = 1 - 2m/r$:

$$\frac{dp_r}{dr} = -\frac{1}{2}(\epsilon + p_r)\nu' + \frac{2}{r}(p_t - p_r).$$

Substituting $\nu' = 2(m + 4\pi r^3 p_r)/[r(r-2m)]$ from the Einstein equations yields the full anisotropic TOV.

9.2 The Anisotropic TOV Equation

Substituting ν' from the Einstein equations:

$$\frac{dp_r}{dr} = -\frac{(\epsilon + p_r)(m + 4\pi r^3 p_r)}{r(r-2m)} + \frac{2}{r}(p_t - p_r). \tag{39}$$

The first term is the standard isotropic TOV. The second term is the **anisotropy force**: positive when $p_t > p_r$ (tangential stresses exceed radial), negative when $p_t < p_r$.

The mass equation $dm/dr = 4\pi r^2 \epsilon$ is unchanged (it follows from the tt Einstein equation, which is insensitive to the pressure anisotropy for a spherically symmetric spacetime).

9.3 Physical Meaning of the Anisotropy Force

Consider the case $p_t > p_r$ (tangential pressure dominates). The term $+(2/r)(p_t - p_r)$ is *positive*, acting like an outward force that partially counteracts gravity. Physically, a large tangential stress means the matter is "squeezed sideways" — the excess tangential pressure supports the star against radial compression.

This has the following consequences:

- **Larger maximum mass**: With extra support from tangential stresses, the star can hold more mass before collapsing.
- **Greater compactness**: The Buchdahl bound $2M/R < 8/9$ (which assumes isotropy) can in principle be exceeded if $p_t > p_r$. Maximally anisotropic stars can

approach the Schwarzschild limit $2M/R \to 1$.

- **Modified tidal response**: The Love number k_2 is altered because the equilibrium configuration changes, which modifies the tidal perturbation equations.

Conversely, $p_t < p_r$ (radial dominates) *reduces* the maximum mass and compactness.

9.4 Bowers–Liang Parameterisation

A widely used phenomenological model for anisotropy is the Bowers–Liang (1974) prescription:

$$p_t - p_r = \frac{\lambda_{\mathrm{BL}}}{3}(\epsilon + 3p_r)(\epsilon + p_r)\frac{r^2}{1 - 2m/r}, \tag{40}$$

where λ_{BL} is a free dimensionless parameter.

- $\lambda_{\mathrm{BL}} = 0$: isotropic (standard TOV).
- $\lambda_{\mathrm{BL}} > 0$: $p_t > p_r$, effectively stiffer.
- $\lambda_{\mathrm{BL}} < 0$: $p_t < p_r$, effectively softer.

This parameterisation has the appealing property that it vanishes at the centre (where $r = 0$) and at the surface (where $\epsilon + 3p_r \to 0$), respecting the boundary conditions.

Anisotropy should be thought of as a correction to the dominant isotropic pressure, not a replacement for the EOS. Very large anisotropy ($|\lambda_{\mathrm{BL}}| \gg 1$) can produce unphysical configurations (e.g. negative energy density or superluminal sound). Realistic anisotropy from magnetic fields or elasticity is a small effect in most stars, and its inclusion in structural modelling is primarily useful for:

1. Testing robustness of mass–radius constraints,
2. Studying extreme hypothetical configurations,
3. Modelling magnetars or crystalline quark stars.

10 Superfluidity in Neutron Star Interiors

10.1 Pairing in Dense Matter

At the densities found in neutron star interiors, nucleons interact via the nuclear force. A key consequence is that neutrons (and protons) form **Cooper pairs**, analogous to the Cooper pairs of electrons in a superconductor.

The pairing gap Δ satisfies (schematically):

$$\Delta \sim E_F \exp\left(-\frac{\pi}{2k_F|a|}\right), \tag{41}$$

where a is the scattering length of the relevant pairing channel.

Here a is the scattering length characterising the interaction between the paired particles. A large $|a|$ corresponds to a strong attractive interaction, leading to a larger pairing gap.

The pairing gap Δ is the minimum energy required to break a Cooper pair and create quasiparticle excitations. It represents a many-body energy scale (not a two-body binding energy) and leads to a gap in the excitation spectrum at the Fermi surface.

The critical temperature below which pairing occurs is

$$T_c \approx \frac{0.57\,\Delta}{k_B}, \tag{42}$$

which is the standard weak-coupling BCS result

$$T_c = \frac{e^\gamma}{\pi}\frac{\Delta}{k_B} \approx 0.567\,\frac{\Delta}{k_B}.$$

Region	**Species**	**Channel**	T_c
Inner crust	Neutrons	$^1S_0$	$\sim 10^9$ K
Outer core	Neutrons	$^3P_2$	$\sim 10^8$–10^9 K
Outer core	Protons	$^1S_0$	$\sim 10^9$–10^{10} K

Since neutron stars cool to $T \lesssim 10^8$ K within $\sim$ 10–100 yr, and below T_c within minutes of formation, **essentially the entire neutron star interior is in the superfluid/superconducting state for most of its lifetime**.

10.2 Pairing Channels

10.2.1 The $^1S_0$ Channel (Singlet Pairing)

At low density ($n \lesssim n_0$), the neutron–neutron scattering length in the $^1S_0$ channel is large and negative ($a_{nn}^{^1S_0} \approx -18.5$ fm), meaning the interaction is attractive. This produces strong singlet pairing in the inner crust.

However, at higher density the $^1S_0$ interaction becomes repulsive (the potential passes through zero at $k_F \sim 1.8$ fm^{-1}). Thus singlet pairing is suppressed in the core.

10.2.2 The $^3P_2$ Channel (Triplet Pairing)

In the core, neutrons pair in the $^3P_2$ channel (spin-triplet, $\ell = 2$, $j = 2$). This channel is attractive at the relevant densities. The triplet gap is significantly smaller than the singlet gap: $\Delta_{^3P_2} \sim 0.1$–1 MeV, giving $T_c \sim 10^8$–10^9 K.

11 Two-Fluid Hydrodynamics

11.1 Motivation

In the superfluid state, the neutron fluid acquires a macroscopic quantum wavefunction and its hydrodynamics differs fundamentally from a normal fluid. The key new feature is the **decoupling** of the superfluid neutron flow from the normal charged component (protons and electrons).

The minimal description requires two velocity fields:

$$u_n^\mu \quad \text{(neutron superfluid four-velocity)}, \qquad u_p^\mu \quad \text{(proton–electron normal component four-velocity)}. \tag{43}$$

11.2 Entrainment

In a superfluid mixture, the momentum of one component is dragged by the current of the other — this is called **entrainment**. The momentum densities are:

$$\pi_\mu^{(n)} = \mathcal{B}^{nn} n_\mu^{(n)} + \mathcal{B}^{np} n_\mu^{(p)}, \tag{44}$$

$$\pi_\mu^{(p)} = \mathcal{B}^{pn} n_\mu^{(n)} + \mathcal{B}^{pp} n_\mu^{(p)}, \tag{45}$$

where $n_\mu^{(i)} = n_i u_\mu^{(i)}$ is the four-current of species i and $\mathcal{B}^{ij}$ are the **entrainment coefficients** (forming a symmetric matrix).

Entrainment means *momentum and mass current are not parallel.* In the absence of entrainment, the momentum of the neutrons would simply be $m_n n_n v_n$. With entrainment, the neutron momentum density also contains a term proportional to the *proton* velocity. Physically this happens because the quasiparticles (Bogoliubov excitations) above the superfluid ground state carry momentum from both species. The entrainment coefficient in the crust is particularly large and has significant consequences for pulsar glitch dynamics.

11.3 Superfluid Equations of Motion

The equations of motion for the two-fluid system are (schematically):

$$\partial_\mu(n_n u_n^\mu) = 0 \quad \text{(neutron baryon conservation)}, \tag{46}$$

$$\partial_\mu(n_p u_p^\mu) = 0 \quad \text{(proton baryon conservation)}, \tag{47}$$

$$u_n^\nu \nabla_{[\nu}\left(\pi_{\mu]}^{(n)}\right) = f_\mu^{(n)} \quad \text{(neutron equations of motion)}, \tag{48}$$

$$u_p^\nu \nabla_{[\nu}\left(\pi_{\mu]}^{(p)}\right) = f_\mu^{(p)} \quad \text{(proton equations of motion)}, \tag{49}$$

where $f_\mu^{(i)}$ are mutual friction and electromagnetic forces. In equilibrium, $u_n^\mu = u_p^\mu$ (co-rotation) and the standard single-fluid TOV is recovered.

12 Relevance for Tidal Physics

12.1 How Each Topic Modifies the Tidal Response

Everything discussed today modifies the tidal response of a neutron star, which determines the gravitational-wave phase in a binary inspiral.

12.1.1 EOS and Phase Transitions

The tidal deformability $\Lambda = (2/3)k_2(R/M)^5$ depends sensitively on R, which is controlled by the EOS. Phase transitions introduce sharp features:

- A **first-order transition** softens the EOS over a density range, reducing R and hence Λ.
- If the transition is strong enough to produce twin stars, a binary could involve two stars at the same mass but wildly different Λ: this would produce a very asymmetric tidal waveform.
- The sound speed jump at a phase boundary can excite **discontinuity g-modes**, which contribute to the dynamical tidal response at frequencies $\sim$ 50–200 Hz.

12.1.2 Anisotropic Pressure

Anisotropy modifies the equilibrium star, and hence the background metric on which the tidal perturbation propagates. The tidal perturbation equations must be solved on the anisotropic background. The Love number k_2 receives corrections from both $p_t - p_r$ and its radial derivative. For moderate anisotropy $|\lambda_{\rm BL}| \lesssim 2$, the correction to k_2 is at the level of $\sim$ 5–20% — potentially observable with next-generation detectors.

12.1.3 Superfluidity and Oscillation Modes

Superfluidity introduces additional oscillation modes absent in a single-fluid star:

- **Superfluid g-modes**: buoyancy-driven oscillations of the superfluid–normal fluid interface, with frequencies $\sim$ 10–100 Hz.
- **Superfluid inertial modes**: Coriolis-driven modes that couple to gravitational waves in rotating stars.

These modes contribute to the **dynamical tidal response**, which becomes important at orbital frequencies above $\sim$ 50 Hz (late inspiral). If a mode frequency matches the orbital frequency, a **resonance** occurs and the tidal energy deposition spikes, leaving a distinctive imprint in the gravitational-wave phase.

The gravitational-wave phase accumulation from a resonant mode is:

$$\Delta\Phi_{\rm res} \sim -\frac{T_{\rm res}}{\mathcal{M}_{\rm chirp}} \int_{f_{\rm res}} \frac{\mathcal{E}_{\rm mode}(f)}{f}\, df, \tag{50}$$

where $\mathcal{E}_{\rm mode}$ is the energy deposited in the mode. For the f-mode (fundamental fluid mode), resonance occurs near merger ($f_{\rm orb} \sim$ 500–1000 Hz) and contributes $\sim$ 1 radian of phase — detectable with next-generation instruments. For the g-modes and superfluid modes, resonance can occur earlier in the inspiral, making them potentially more accessible with current detectors at design sensitivity. Here $T_{\rm res}$ is the effective duration over which the system remains in resonance (of order the inverse frequency sweep rate), so this expression is schematic and intended only as an order-of-magnitude estimate.

13 The Neutron Star Crust: Structure and Phases

The crust of a neutron star is a thin ($\sim$ 1 km) but structurally rich region, with distinct phases from surface to core.

Region	Density	Structure	Physics
Outer crust	$\rho < 4 \times 10^{11}$ g/cm^3	Lattice of nuclei + electrons	Baym–Pethick–Sutherland E
Inner crust	4×10^{11}–2×10^{14} g/cm^3	Nuclei + dripped neutrons + e^-	Neutron superfluidity
Nuclear pasta	$\sim 0.5\,\rho_{\rm sat}$	Rods, slabs, bubbles, tubes	Frustrated geometry
Outer core	$\rho_{\rm sat}$–$2\rho_{\rm sat}$	Uniform $npe\mu$ matter	$^3P_2$ superfluidity
Inner core	$> 2\rho_{\rm sat}$	Unknown (hyperons? quarks?)	Open question

13.1 Nuclear Pasta

At densities $\sim$ 0.3–0.7 $\rho_{\rm sat}$, the competition between nuclear surface tension and Coulomb repulsion produces a sequence of exotic nuclear shapes known as **nuclear pasta**: spherical bubbles $\to$ cylindrical tubes $\to$ slabs $\to$ cylindrical rods $\to$ spherical nuclei (as density decreases).

Nuclear pasta is particularly relevant for:

- **Shear modulus**: Pasta phases have complex elastic properties that affect the crust's response to deformations.
- **Electron transport**: Pasta structures scatter electrons differently from the regular lattice, affecting electrical conductivity and hence magnetic field evolution.

- **Gravitational-wave emission**: Non-axisymmetric pasta deformations ("mountains") in rotating NS could be a continuous GW source — currently constrained to $\epsilon_{\max} \lesssim 10^{-6}$.

14 Summary of the EOS Landscape

The neutron star EOS can be organised by density regime, with different physics dominating in each:

Density	Matter	EOS model	Key observable
$n < 0.5\,n_0$	Nuclei, dripped neutrons	Nuclear mass data + theory	Crust thickness, I_{crust}
0.5–$2\,n_0$	Uniform $npe\mu$	Symmetry energy, J, L	Radius $R_{1.4}$
2–$5\,n_0$	Uncertain; possibly hyperons, Δ baryons	Many-body theory, chiral EFT	Maximum mass $M_{\max}$
$> 5\,n_0$	Possibly quark matter	QCD (perturbative at $>40\,n_0$)	$M_{\max}$, post-merger signal

The combined constraints from radio pulsars, NICER, and GW170817 bracket the neutron star EOS to:

- $M_{\max} > 2.0\,M_\odot$ (from massive pulsars),
- $R_{1.4} \approx 11.5$–13.5 km (from NICER + GW170817),
- $\tilde{\Lambda}(1.4\,M_\odot) \lesssim 800$ (from GW170817),
- $c_s^2 > 1/3$ at $n \sim 2$–$4\,n_0$ (from $M_{\max}$ constraint + causality).

These constraints already rule out: purely soft EOS (too low $M_{\max}$), purely stiff EOS (too large Λ), and many hyperon/quark models without repulsive interactions.

15 Conceptual Summary

Today we have extended the Day 1 framework in three directions.

Part I: The Dense Matter EOS

1. Cold catalyzed neutron star matter is in beta equilibrium and charge neutrality, with the thermodynamic state fully determined by the baryon density n.
2. The speed of sound $c_s^2 = dp/d\epsilon$ measures EOS stiffness, bounded by $0 < c_s^2 \leq 1$.
3. The symmetry energy slope L is the nuclear parameter most strongly correlated with neutron star radius.
4. First-order phase transitions soften the EOS and can produce twin-star solutions with two stable branches in the M–R plane.

5. The sound speed likely exceeds $c_s^2 > 1/3$ at intermediate densities, in tension with the conformal limit and hinting at strong repulsive correlations or quark onset.

Part II: Anisotropic Pressure

1. The anisotropic TOV equation adds an effective force $+(2/r)(p_t - p_r)$ to the standard TOV.
2. When $p_t > p_r$, tangential stresses partially counteract gravity, allowing larger masses and compactnesses.
3. The Buchdahl bound $2M/R < 8/9$ can be circumvented with sufficient anisotropy.
4. Anisotropy modifies the equilibrium structure and hence the tidal Love number k_2.

Part III: Superfluidity

1. Neutrons pair in the ${}^1S_0$ (crust) and ${}^3P_2$ (core) channels; protons pair in ${}^1S_0$ and form a type-II superconductor.
2. The two-fluid description requires entrainment between the superfluid neutron current and the normal proton–electron current.
3. Quantised vortices carry angular momentum; vortex pinning and unpinning is the mechanism for pulsar glitches.
4. Mutual friction couples the two fluids on timescales of days to years, visible in post-glitch recovery.
5. Superfluidity introduces new oscillation modes that contribute to the dynamical tidal response.

16 Exercises

Exercise 16.1. Starting from the Euler relation for a thermodynamic system at $T = 0$, derive the relation $p = n\mu - \epsilon$. Then show that $dp = n\, d\mu$ (Gibbs–Duhem relation at $T = 0$) and use it to derive the expression for c_s^2.

Exercise 16.2. For an ultra-relativistic free Fermi gas, show explicitly that:

(a) $p = \epsilon/3$,

(b) $c_s^2 = 1/3$,

(c) The EOS corresponds to a polytrope with $\Gamma = 4/3$.

Exercise 16.3. Derive the chemical equilibrium condition for beta equilibrium in a system containing n, p, e^-, and μ^-. Under what conditions does the direct URCA process operate, and what is the threshold proton fraction?

Exercise 16.4. Show explicitly that the anisotropic TOV equation (39) reduces to the standard isotropic TOV when $p_r = p_t$. Then verify that the mass equation $dm/dr = 4\pi r^2 \epsilon$ is unchanged by anisotropy.

Exercise 16.5. For the Bowers–Liang parameterisation, compute $p_t - p_r$ at the stellar centre (show it vanishes) and at an interior point $r > 0$. Explain physically why the anisotropy vanishes at the centre even for $\lambda_{\rm BL} \neq 0$.

Exercise 16.6 (Advanced)**.** The PREX-II experiment measured the neutron skin thickness of ^{208}Pb as $\Delta r_{np} = 0.283 \pm 0.071$ fm, giving $L \approx 106 \pm 37$ MeV. Using the approximate correlation $R_{1.4} \approx 11 + 0.5\left(\frac{L}{100\,\mathrm{MeV}}\right)^{1/4}$ km, estimate the radius implied. Compare with the NICER constraint $R_{1.4} = 11.5$–13.5 km. What are the possible resolutions of the tension?

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